

Assignment 1 - Part B

CS 329E - Elements of Data Analytics
Relational Algebra
(20 Points)

Running Example: University Course Registration Database

Consider a university dataset with the following relations:

- **Student**(SID, Name, Major, Year)
- **Course**(CID, Title, Department, Credits)
- **Instructor**(IID, Name, Department)
- **Enrollment**(SID, CID, Grade)
- **Teaches**(IID, CID, Semester)

Primary keys are underlined. The **Enrollment** relation represents students registered in courses, and **Teaches** represents which instructor teaches which course in a given semester.

Example Records

Student

SID	Name	Major	Year
S001	Alice Johnson	CS	2
S002	Bob Smith	Math	3
S003	Carol Lee	CS	4
S004	David Kim	Physics	1
S005	Eva Brown	CS	3
S006	Frank Miller	Biology	2
S007	Grace Wilson	CS	1

Course

CID	Title	Department	Credits
CS101	Intro to Computer Science	CS	4
CS201	Data Structures	CS	4
CS301	Databases	CS	3
MATH101	Calculus I	Math	4
PHYS101	General Physics	Physics	4
BIO101	Introduction to Biology	Biology	3

Instructor

IID	Name	Department
I001	Prof. Anderson	CS
I002	Prof. Martinez	Math
I003	Prof. Chen	Physics
I004	Prof. Gupta	CS
I005	Prof. Robinson	Biology

Enrollment

SID	CID	Grade
S001	CS101	A
S001	CS201	B
S002	MATH101	A
S003	CS101	A
S003	CS301	A
S004	PHYS101	B
S005	CS201	C
S005	CS301	B
S007	CS101	A

Teaches

IID	CID	Semester
I001	CS101	Fall 2024
I004	CS201	Fall 2024
I004	CS301	Spring 2025
I002	MATH101	Fall 2024
I003	PHYS101	Fall 2024
I005	BIO101	Spring 2025

Tasks

Develop Relation Algebra for the following Queries:

Query 1: Find the names of all students majoring in Computer Science

Query 2: Find the titles of all courses offered by the Computer Science department

Query 3: Find the names of students enrolled in the course with CID = 'CS101'

Query 4: Find the names of instructors who teach CS courses

Query 5: Find students who are enrolled in at least one course taught in Fall 2024

Query 6: Find students who are enrolled in all CS courses

Query 7: Find students who are *not* enrolled in any course

Query 8: Find courses that have no enrolled students

Query 9: Find students who received an 'A' grade in a CS course

Query 10: Find instructors who teach more than one course